pve-fcsan: How It Works

Version 1.4.0

Goals and the stack mapping

pvesan rebuilds the vCenter SAN workflow on top of parts Proxmox VE already trusts: dm-multipath for paths, shared thick LVM for the datastore, and pmxcfs for cluster state. It adds no on-disk format and no kernel code. If you remove pvesan, the datastores are still ordinary LVM storages.

Goals, in priority order

1. Never lose or corrupt data. Every destructive step checks usage on every node first, and LVM metadata changes only happen under Proxmox VE's own cluster storage lock.
2. Match the vCenter operator experience: the same verbs, run cluster-wide from any node, with a task log.
3. Stay upgrade-safe. Change as little of pve-manager as possible, and re-apply those changes automatically after an upgrade.
4. Keep GUI, CLI and API identical, by building the CLI and GUI on the API.

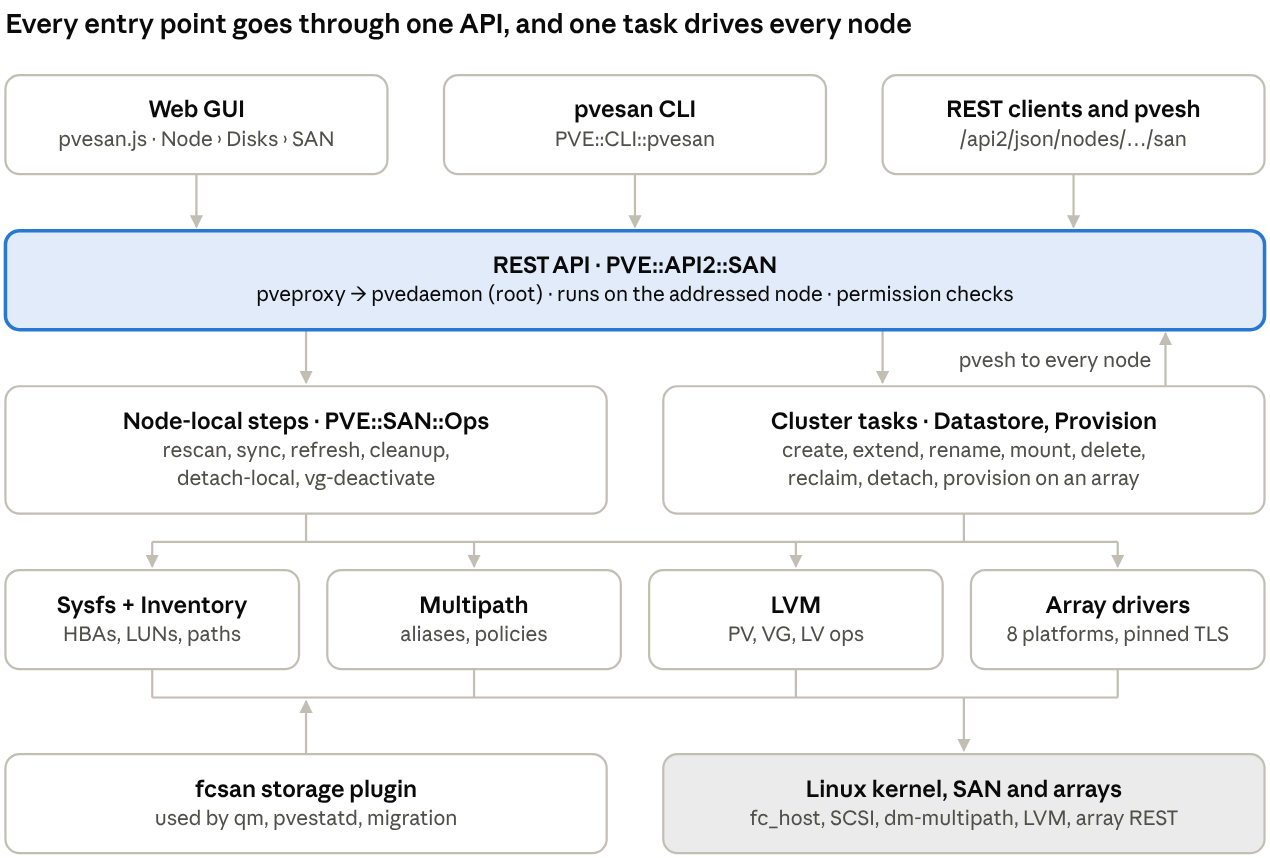
Layer	ESXi / vCenter	Proxmox VE + pvesan
HBA and transport	vmhba, FC driver	`qla2xxx` / `lpfc`, `/sys/class/fc_host`
Device identity	`naa.` device ID	SCSI WWID (`3` + NAA hex), same value
Multipathing	NMP with SATP and PSP	dm-multipath; the policy maps onto `path_grouping_policy`, `prio` and `failback`
Device display name	Device display name	Multipath `alias` (`/dev/mapper/<alias>`), kept cluster-wide
Datastore file system	VMFS, a clustered FS with on-disk locks	Shared thick LVM VG; metadata serialized by the pmxcfs cluster lock
VM disk	VMDK file on VMFS	One LV per disk (raw, or a qcow2 volume chain on Proxmox VE 9)

Layer	ESXi / vCenter	Proxmox VE + pvesan
Datastore registry	vCenter inventory	`/etc/pve/storage.cfg` entry of type `fcsan`
Cluster-wide config	vCenter DB, host profiles	`/etc/pve/pvesan/config.json` on pmxcfs
Rescan	`esxcli storage core adapter rescan`	sysfs `scan` and `rescan`, then a multipathd resize
Detached device	Device state `off`	WWID blacklisted, paths deleted, removed again after each rescan
Array plugin (Pure, InfiniBox, ...)	vSphere plugin	pvesan array drivers

The largest difference is the datastore layer. VMFS lets many hosts allocate space at the same time using on-disk locks. Shared LVM has one metadata writer at a time, enforced with a cluster-wide lock, while data I/O to existing LVs runs in parallel on every node. So the design rule is: metadata changes (create, extend, resize, delete) go through the lock, data paths never do.

Architecture

The GUI, the CLI and REST clients all call one API. A cluster task runs on the node it was called on and reaches every other node through that same API. The storage plugin is the only piece Proxmox VE itself calls, from `qm`, `pvestatd` and migration.



pvesan architecture · entry points, API, orchestration, backends

Component	File	Integration point
Storage type `fcsan`	`PVE/Storage/Custom/FCSANPlugin.pm`	Proxmox VE loads every module in `Storage/Custom`. It subclasses `LVMPlugin` and overrides `activate_storage`, `options` and `properties`
REST API	`PVE/API2/SAN.pm`	Registered as `/nodes/{node}/san` by one marked line in `PVE/API2/Nodes.pm`
CLI	`PVE/CLI/pvesan.pm`, `/usr/sbin/pvesan`	`PVE::CLIHandler` maps each command onto an API method

Component	File	Integration point
GUI	`/usr/share/pve-manager/js/pvesan.js`	One `<script>` line in `index.html.tpl`; an Ext `override` of `PVE.panel.Config` adds the panel to the node's Disks group
Integration	`/usr/share/pve-fcsan/pvesan-integrate`	Run by `postinst`, re-run by a dpkg trigger when pve-manager replaces either file
Boot sync	`pvesan-sync.service`	`pvesan sync` after `pve-cluster` and `multipathd`, before `pvestatd` and guests
Array drivers	`PVE/SAN/Array/*.pm`	Called only by provisioning workflows

Data model

State lives in three places, each with one owner; each node renders a fourth from them.

Store	Location	Holds	Written by
Storage registry	`/etc/pve/storage.cfg` (pmxcfs)	`fcsan` entry: `vgname`, `shared`, `nodes`, `disable`, `fc-wwids`, `fc-autorescan`	Proxmox VE storage API (`PVE::API2::Storage::Config`)

Store	Location	Holds	Written by
pvesan cluster DB	`/etc/pve/pvesan/config.json` (pmxcfs)	Per WWID: `alias`, `policy`, `detached`, `datastore`. Per datastore: `label`, linked `array` and `volumes`. Per array: type, address, user, fingerprint, vendor settings	`PVE::SAN::Config::update` under the `pvesan` cfs domain lock
Array secrets	`/etc/pve/priv/pvesan/<array>.pw`	Password or API token	Root only
LVM metadata	On the LUNs	VG, PVs, LVs	LVM commands, only under `cluster_lock_storage`
Rendered, per node	`/etc/multipath/conf.d/pvesan.conf`, `/etc/multipath/wwids`, LVM devices file	Aliases, policies, blacklist of detached WWIDs, claimed WWIDs	`PVE::SAN::Ops::sync`

Why a separate cluster DB? Aliases and path policies apply to LUNs, not storages: a LUN can have a policy before it belongs to a datastore, and a detached LUN belongs to none. `storage.cfg` has no place for that, so pvesan keeps a small JSON file next to it on pmxcfs.

Rendering. `render_conf()` is a pure function (device table in, multipath snippet out, sorted by WWID), so the same input always gives the same bytes. `sync` writes the file and runs `multipathd reconfigure` only when the content changed, which makes it cheap and safe to run at boot, during activation, and twice in every create.

```
multipaths {
        multipath {
                wwid 3600a098038303053453f463045727a41
                alias ds1
                path_grouping_policy group_by_prio
                prio alua
                path_selector "service-time 0"
                failback immediate
        }
}
```

Identity. Everything is keyed by WWID, never by `sdX` or `dm-N`, which change across reboots and differ between nodes. sysfs gives `naa.<hex>`; pvesan turns it into multipath's form, `3<hex>`. For vendor-specific (t10) IDs it calls `scsi_id`, the same source multipath uses.

Usage classification. For each LUN, `Inventory::devices` decides one usage, in this order: `datastore:<id>`, `lvm-vg:<vg>`, `lvm-pv`, `in-use:<holders>`, `signature:<type>` / `partitioned:<type>` (from blkid), `free`. Only `free` LUNs can be used without `force`; the first four are never wiped.

Operation internals

Every cluster operation follows the same pattern:

1. Check the situation on every node through the API.
2. Change cluster state on pmxcfs.
3. Apply that state on every node.
4. Make the single LVM metadata change on the coordinating node, under the lock.

The fan-out (`PVE::SAN::Cluster::on_nodes`) calls `pvesh <verb> /nodes/<n>/san/<step>` for each node, the local one included, so every node runs the same code. It collects every node's failure before it aborts, so one task log shows all problems.

Rescan (`Ops::rescan`). Snapshot the SAN paths; optionally issue a LIP per port and wait for `Online`; write `- - -` to each FC/iSCSI host's `scan`; write `1` to each known path's `rescan` to re-read capacity; `udevadm settle`; delete paths of detached WWIDs; `multipathd resize map` for maps whose paths grew; return the difference.

Sync (`Ops::sync`). Render the snippet, write it if changed, and add or remove WWIDs in `/etc/multipath/wwids`. pvesan edits that file directly instead of calling `multipath -a`, which needs the LUN to be visible, and not every node sees every LUN. If anything changed, run `multipathd reconfigure` and wait for the maps of LUNs this node has paths to; add them to the LVM devices file when it is enabled.

Create (`Datastore::create`).

1. Refuse if the storage ID or VG exists.
2. `check_visible`: every target node must see each WWID with an active path.
3. `check_unused_locally`: classify each LUN.
4. `register_devices`: alias in the cluster DB, `sync` on all nodes, wait for `/dev/mapper/<alias>`.
5. `wipefs` (only with `force`), `pvcreate`, `vgcreate --addtag pvesan`.
6. `PVE::API2::Storage::Config->create(type => 'fcsan', shared => 1, ...)`, the same call Proxmox VE's own Disks → LVM wizard makes.
7. Roll back with `vgremove` / `pvremove` if step 5 or 6 fails; finally `sync` everywhere again.

Extend. *Grow* runs `devices/{wwid}/refresh` on every node (re-read size, resize map) and aborts if the nodes disagree; only then `pvresize` under `cluster_lock_storage`. *Add extent* reuses steps 2–5 of create, then `vgextend` under the lock and updates `fc-wwids`.

Rename. A label rename touches only the cluster DB. An ID rename requires every node online and no running guest that refers to the storage, deactivates the VG everywhere, moves the `storage.cfg` section (and optionally runs `vgrename`) inside `lock_storage_config`, then rewrites `^key: <id>:` references in every guest config, snapshot sections included.

Unmount and mount map onto Proxmox VE's two switches: removing nodes from `nodes`, or `disable 1`. The VG is deactivated on the affected nodes first, and `vg-deactivate` refuses if any LV is open.

Delete refuses while guest configs refer to the storage or LVs remain (without `destroy-data`), deactivates everywhere, deletes the storage through the API, runs `vgremove` / `pvremove` / `wipefs`, cleans the cluster DB and optionally detaches the LUNs.

Detach asks every node whether the LUN is in use, marks it `detached` (which blacklists and unclaims it on `sync`), then runs `detach-local` on every node to flush the map and delete each path. Every later rescan deletes the paths again.

Reclaim checks `discard_max_bytes > 0` on each backing map, creates `pvesan_reclaim` over the free space minus a reserve under the storage lock, releases the lock, runs `blkdiscard`, and removes the LV again under the lock.

Activation (`FCSANPlugin::activate_storage`). Before handing over to `LVMPlugin`, the plugin checks that each WWID in `fc-wwids` has a map. If one is missing and `fc-autorescan` is on, it runs a local rescan and sync, rate-limited to once per 5 minutes per storage by a stamp file in `/run/pvesan`. If the map is still missing, activation fails with a clear message and the storage shows inactive on that node only.

Array integration

Array drivers turn pvesan from a SAN manager into a provisioning tool. The array becomes one more layer pvesan drives, ahead of the steps above.

Module	Role
`PVE::SAN::Array`	Registry of array types to drivers; array settings in the cluster DB, secrets in `/etc/pve/priv/pvesan/`; certificate probe; collects every node's WWPNs
`PVE::SAN::Array::Base`	Shared driver code: TLS-pinned HTTP client, naming rules, vendor-neutral host and host-cluster registration
`PVE::SAN::Array::<Vendor>`	One module per platform
`PVE::SAN::Provision`	Workflows: register the cluster, provision, grow, destroy with the volume, link

Driver contract. A driver implements `system_info`, `pools`, `hosts`, `host_clusters`, `create_hostcluster`, `create_host`, `add_host_ports`, `add_to_hostcluster`, `create_volume`, `volume`, `find_volume_by_wwid`, `map_to_hostcluster`, `unmap_from_hostcluster`, `expand_volume` and `delete_volume`. A volume always comes back with its multipath WWID, which connects array objects to everything else in pvesan. Drivers whose arrays present volumes differently override `ensure_hostcluster`. Three hooks cover vendor differences: `options` (required settings such as `svm` or `portgroup`), `valid_hostcluster_name`, and `wwid_after_map`.

Trust. No connection is made without a pinned SHA-256 certificate fingerprint. `array-probe` reads it, the admin compares it with the array's own display, and from then on only that certificate is accepted. Hostname checks are off because the pin is stronger than CA trust for the usual self-signed array certificate. Credentials are tested before anything is saved.

Registration (`ensure_hostcluster`) is idempotent. For each node, an array host that already owns any of its WWPNs is reused and missing ports are added; otherwise a host named after the node is created (with "force", because ports may not be logged in until zoning is done). Every host then joins the cluster's host cluster. Ports split across hosts, or a host already in another host cluster, are refused: fixing those is an admin decision on the array.

Provision runs inside one task: register the cluster; create the volume; map it; rescan every 30 s and poll every node every 10 s until each node has an active path (up to 120 s); record the path policy; run the normal datastore create; store the link `datastores.<id> = { array, volumes[] }`. If anything fails before the datastore exists, the task unmaps the volume, cleans dead paths on all nodes, deletes the volume and drops the device record.

Platform	API	Checked against	How the cluster is presented	Volume ID → WWID
IBM Spectrum Virtualize	`POST /rest/v1/<cli-command>[/<obj>]`, token from `/auth`	IBM REST API reference	Host cluster	`3` + `vdisk_UID`
Pure FlashArray	`/api/2.x/...`, API token → `x-auth-token`	Pure's generated SDK docs (go-pure-client)	Host group; mapping = connection	`3624a9370` + serial
INFINIDAT InfiniBox	`/api/rest/...`, Basic auth, `approved=true` on changes	InfiniSDK source	Cluster; LUN on the cluster, unmapped by LUN number	`36` + serial
Dell PowerStore	`/api/rest/...`, `login_session` → `DELL-EMC-TOKEN` + cookie	PyPowerStore	Host group (created with its first host)	`wwn`
Dell PowerMax / VMAX	Unisphere `/univmax/restapi/<ver>/sloprovisioning/symmetrix/<id>`	PyU4V	Host group + storage group `<hc>_SG` + masking view `<hc>_MV`; volumes created in the SG	`effective_wwn`
NetApp ONTAP	`/api/...`, jobs awaited	`netapp.ontap` Ansible modules	One FCP igroup; one FlexVol + LUN per datastore	`3600a0980` + hex(serial)

Platform	API	Checked against	How the cluster is presented	Volume ID → WWID
NetApp E-Series	`/devmgr/v2/storage-systems/<ssid>`	SANtricity Ansible collection	Host group (Linux DM-MP host type)	`wwn`
Hitachi VSP	`/ConfigurationManager/v1/objects/storages/<id>`, session token, jobs awaited	OpenStack Cinder Hitachi driver, Configuration Manager reference	One host group per target port; LU path on each	`naaId` (known after mapping)
Fujitsu ETERNUS DX S5 / AF S3	`:5665/api/v1/...`, `POST session` → Bearer token, jobs awaited and deleted	ETERNUS RESTful API Reference (P3AG-5492)	One FC host per WWN (the API has no host groups); every volume connected to all of them with one host LUN	`3` + `uid`

Consistency, locking, safety, permissions

The one invariant: **only one node at a time writes LVM metadata for a shared VG, and every node sees the same device size before LVM is told about it.**

Lock	Taken through	Guards	Held for
Cluster storage lock	`PVE::Storage::Plugin->cluster_lock_storage($id, 1, 60, ...)`	`pvcreate` / `vgextend`, `pvresize`, reclaim LV create/remove; Proxmox VE's own volume allocation takes it too	Seconds
storage.cfg lock	`PVE::Storage::lock_storage_config`	ID rename (section move plus `vgrename`)	Seconds
pvesan domain lock	`cfs_lock_domain('pvesan')`	Read-modify-write of `config.json`	Milliseconds

pmxcfs aborts code holding a lock for more than 60 s, so every slow step (node fan-out, waiting for maps, `blkdiscard`, array calls) stays outside locks.

Safety rails

- Visibility gate: create and extend abort unless every target node sees each LUN; grow aborts if nodes report different sizes.
- Usage gate: LUNs in use as a datastore, VG, PV or with holders are never wiped; signatures and partitions need explicit `force`.

- Reference gate: delete, ID rename and unmount refuse while guest configs refer to the storage or those guests run on the target nodes.
- Open-volume gate: `vg-deactivate` refuses while any LV is open.
- Rollback: a failed create removes the VG/PVs it made; a failed provision deletes the new array volume.
- Typed confirmation for Delete and Detach in the GUI.
- Integration guard: the line in `Nodes.pm` loads pve-fcsan inside an `eval`, so a pve-fcsan that fails to load disables only `/nodes/{node}/san` (with a warning in the journal), never the rest of the API. `pvesan-integrate` runs `perl -c` on the patched `Nodes.pm` and reverts if it fails. The GUI override catches its own errors, so a broken `pvesan.js` loses only the SAN panel.
- Uninstall guard: `prerm` refuses while `fcsan` storages exist.

Permissions. API calls are `protected` (handled by pvedaemon as root) and `proxyto => 'node'`. Reads need `Sys.Audit` on the node; node-local changes `Sys.Modify` on the node; anything cluster-wide or array-related `Sys.Modify` on `/` and `Datastore.Allocate` on `/storage`. Those privileges allow root-level storage changes on every node, the same as Proxmox VE's own storage management.

Development and testing

Design decisions

Decision	Chosen	Rejected, and why
Datastore format	Shared thick LVM (`fcsan` subclasses `LVMPlugin`)	GFS2/OCFS2: extra cluster stack, not supported by Proxmox. LVM-thin: not safe shared
Storage type	Custom `fcsan` type	Plain `lvm`: no LUN binding or auto-rescan (plain `lvm` storages on SAN are still listed and managed)
API exposure	One line in `Nodes.pm`, re-applied by a dpkg trigger	A separate daemon: second auth path and port. Runtime monkey-patching: fragile load order
GUI	Ext `override` of `PVE.panel.Config`	Patching `pvemanagerlib.js`: replaced on every update

Decision	Chosen	Rejected, and why
Cluster fan-out	`pvesh` to each node, including the local one	Direct local calls: two code paths to test
Claiming LUNs	Edit `/etc/multipath/wwids` directly	`multipath -a`: fails on nodes that don't see the LUN
Plugin API version	Return the running `APIVER`	Pinning a range: a release for every Proxmox VE bump
Array drivers	Thin per-vendor modules on a shared base	One generic SMI-S driver: patchy vendor support, heavier dependencies

Layers. `Util`, `Sysfs`, `Multipath` (parsing/rendering) and `LVM` (parsing) are pure Perl and read sysfs through `PVESAN_SYSFS_ROOT`, so a fixture tree replaces `/sys`. `Inventory`, `Ops`, `Config` and `Cluster` add Proxmox VE and command execution; `Datastore` and `Provision` compose them; `API2::SAN` exposes them; the CLI and GUI sit only on the API.

Tests (`make test`, 356 checks):

File	Covers
`t/10-util.t`	WWN/WWID normalization, aliases
`t/20-sysfs.t`	Discovery against a fake sysfs tree: FC and iSCSI hosts, paths, maps, ALUA standby, degraded LUNs, grown LUNs
`t/30-multipath.t`	Path-state parsing, golden multipath config render
`t/40-lvm.t`	LVM JSON reports
`t/50-cluster.t`	The exact `pvesh` command lines of the fan-out
`t/60…66-*.t`	Every array driver against a fake array that keeps state (`t/lib/FakeArray.pm`): auth, request bodies, registration, full volume lifecycle, error formats

File	Covers
`t/80-integrate.t`	`pvesan-integrate` against a fake pve-manager tree: insertion point, upgrade from an older line, refusal without the anchor, exact removal, and that the patched `Nodes.pm` still loads with a broken pve-fcsan
`t/90-compile.t`	Loads every module; fails on compile warnings. Uses the real Proxmox VE libraries when present, stubs otherwise

The build also checks that every Proxmox function the code calls exists in the installed libraries, that the REST router accepts all 36 API paths, that `pvesan.js` parses, and that every module is in the package. CI runs the tests on every push.

Validation so far. The plugin runs on a 6-node Proxmox VE 9.2 cluster with QLogic and Emulex HBAs: install, integration, services, storage plugin, API, CLI, GUI asset, local and cluster rescan, refusal of invisible LUNs and CLI exit codes are verified there, and every array driver's first call was checked over pinned TLS. That lab work found and fixed seven bugs that unit tests could not see (an API path clash, a non-existent helper call, arguments lost in the cluster fan-out, a packaging omission, among others). Array drivers have not yet run against real arrays.

Release steps

1. Update `debian/changelog` and `CHANGELOG.md`.
2. `make test` and `make deb` on a clean Debian 12/13 host (CI does this on every tag).
3. Run the cluster acceptance checklist below on a lab cluster.
4. Tag `vX.Y.Z`; CI attaches the `.deb` to the GitHub release and rebuilds the signed apt repository on GitHub Pages from every release's `.deb` (`tools/build-apt-repo.sh`, key in the `APT_SIGNING_KEY` secret).
5. Roll out node by node, checking `pvesan-integrate status` after each.

Cluster acceptance checklist

- [ ] Present a LUN, `rescan-all`, confirm it shows `free` on every node
- [ ] Create a datastore and live-migrate a VM between all nodes
- [ ] Grow the LUN, `extend`, check the size in `pvesm status` on every node
- [ ] Add an extent, reboot one node, check the datastore comes back `active`

- [ ] Set policy `alua`, check `multipath -ll`, pull a fabric cable during I/O
- [ ] Rename the label online; rename the ID with guests stopped and boot them
- [ ] Unmount from one node, check migration to it is refused, mount again
- [ ] Reclaim on a thin LUN and confirm space returns on the array
- [ ] Delete with `--detach`, unmap, rescan, confirm no dead paths remain
- [ ] For each array driver: add, register, provision, grow, delete with volume
- [ ] Upgrade pve-manager and confirm `pvesan-integrate status`

Limitations, known risks, roadmap

Known risks

Risk	Why it might bite	How to check
Array API differences by code level	Drivers were checked against vendor SDKs, not yet against real arrays	Provision and delete a test datastore per array type; report results
Code in pve-storage or qemu-server that tests `type eq 'lvm'`	An `fcsan` storage could miss an LVM-specific branch	Snapshot, clone and migrate a VM on `fcsan`; fall back by changing the type to `lvm`
`Nodes.pm` anchor text changes	The integration can't find its insertion point	`pvesan-integrate apply` reports it and changes nothing
Widget-toolkit API drift	GUI classes could change in a new Proxmox VE	Open the panel after upgrades; check the browser console

Functional limits compared with vCenter

- Thick LVM only at the datastore layer (the array can still be thin). Per-VM snapshots on shared storage need Proxmox VE 9 volume chains.
- No automatic resignaturing of snapshot or replica LUNs (manual `vgimportclone` procedure in the runbook).
- No Storage I/O Control, datastore clusters, Storage DRS or datastore browser. Datastores can't be shrunk.
- Nodes offline during a change catch up at boot through `pvesan-sync.service`.

Roadmap

1. Real-array validation for every driver; fix what it finds.
2. A datacenter-level GUI view: all nodes' paths for one LUN side by side.
3. `datastore adopt`: convert existing `lvm` storages on SAN to `fcsan`.
4. Guided snapshot-LUN import (resignature) and array snapshots.
5. Per-LUN path statistics and fabric events.